Assignment – 03

Hotel Room Booking Form (Advanced) — Advanced Fields Use Case

1. Objective

To extend the existing "Hotel Room Booking Form" application in Zoho Creator by incorporating all seventeen Advanced Fields available in the Form Builder, so that the form moves beyond basic data capture and demonstrates automation, computed values, file/media handling, and structured lookups within a real hospitality-industry use case.

2. Problem Statement

The basic Room Booking Form (built earlier using the thirteen Basic Fields) captures a guest's identity, contact details, and room/stay preferences, but it cannot handle several real-world requirements of a growing hotel property: verifying room-type availability and tariff against a live inventory list, booking rooms for an entire family or group in one submission, collecting advance payment and computing the balance amount after loyalty discounts, issuing a unique scannable booking confirmation, or accepting supporting media such as an ID proof scan, a celebration request video, or a signed terms acknowledgement.
Extend the Room_Booking_Form in Zoho Creator so that, in addition to the existing basic fields, it uses every Advanced Field type provided by Zoho Creator's Form Builder to support group booking, fee computation, room inventory verification, and automated booking-ID/QR generation for a smooth check-in experience.

3. Scope of the Use Case

- Application Name: Hotel Room Booking & Front Office Management (Advanced Edition)
- Form Name: Room_Booking_Form (extended with Advanced Fields)
- Primary Users: Guests / Hotel Front Office staff
- Report View: Room_Booking_Report (extended with computed tariff, booking ID, and QR code columns)
- Objective of the Extension: Enable the front office to verify room inventory automatically, book for families/groups in a single submission, compute payable balances, and issue scannable digital booking confirmations

4. Advanced Fields Used and Their Mapping

The table below maps every Advanced Field from the Zoho Creator Form Builder palette to its role in the extended Hotel Room Booking Form.

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Decision Box	Requires_Airport_Pickup	Captures a quick Yes/No decision on whether the guest needs airport pickup service arranged	Yes / No toggle
Rich Text	Hotel_Policies_Display	Displays formatted hotel policies (bold headings, bullet points, hyperlinks) directly within the form	"Check-in from 1 PM. Check-out by 11 AM. Free cancellation up to 48 hours before arrival."

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
URL	Travel_Itinerary_URL	Captures a link to the guest's travel itinerary or flight booking confirmation page	https://itinerary.example.com/trip/58231
Image	Guest_Photo	Captures a photo of the primary guest for front-desk identity verification at check-in	guest_photo.jpg
Percent	Loyalty_Discount_Percent	Captures the discount percentage applicable for returning or loyalty-program guests	12%
Currency	Advance_Payment_Paid	Captures the advance payment amount paid by the guest at the time of booking, formatted with currency symbol	₹2,000.00
Decimal	Distance_From_Airport_KM	Captures the distance from the airport to the hotel, used to plan the pickup service	14.5
Date-Time	Booking_Submitted_On	Automatically records the exact date and time the booking request was submitted	24-Jul-2026 09:47
SubForm	Additional_Guests_Details	Captures repeating rows of co-traveler name, age, and ID type for group/family bookings	Row 1: Divya A. – 30 – Passport; Row 2: Kiran A. – 6 – School ID
Lookup	Room_Type_Availability_Lookup	Looks up and auto-fills the room tariff and availability status from a master Room Inventory table	Deluxe Double Room – ₹3,200/night – Available
Signature	Guest_Digital_Signature	Captures the guest's digital signature acknowledging the hotel's terms, conditions, and cancellation policy	Signed (drawn signature captured)
File Upload	ID_Proof_Upload	Allows the guest to upload a scanned copy of a valid government ID for check-in verification	guest_id_proof.pdf

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Audio	Special_Request_Voice_Note	Allows the guest to optionally record a short voice note describing any special arrangements needed	voice_note_03.mp3 (0:15)
Video	Celebration_Request_Video	Allows the guest to upload a short video message when requesting room decoration for an anniversary or birthday	celebration_request.mp4 (0:30)
Auto Number	Booking_ID	Automatically generates a unique, sequential booking reference number for every reservation	HTL-BKG-2026-000512
Formula	Total_Amount_Payable	Automatically computes the balance amount due at check-in using the room tariff, discount, and advance paid	= (Room_Tariff × Nights × (1 – Loyalty_Discount_Percent)) – Advance_Payment_Paid
QR/Barcode	Booking_Confirmation_QR_Code	Automatically generates a scannable QR code encoding the Booking ID for quick check-in at the front desk	QR image encoding HTL-BKG-2026-000512

5. Extended Form Layout (Section-wise)

The sections below are added to the existing Guest, Contact, and Room/Stay Preference sections of the basic Hotel Room Booking Form.

Section E — Room Verification & Identity

- Room Type Availability Lookup (Lookup field, auto-fills from master Room Inventory table)
- Guest Photo (Image field)
- ID Proof Upload (File Upload field)
- Distance From Airport (Decimal field)

Section F — Group Booking & Media Details

- Additional Guests Details (SubForm field, repeating rows)
- Travel Itinerary URL (URL field)
- Celebration Request Video (Video field, optional)
- Special Request Voice Note (Audio field, optional)

Section G — Payment & Pickup Preference

- Requires Airport Pickup (Decision Box)
- Advance Payment Paid (Currency field)
- Loyalty Discount Percent (Percent field)
- Total Amount Payable (Formula field, read-only, auto-computed)

Section H — Declaration & System-Generated Values

- Hotel Policies Display (Rich Text field, read-only reference block)
- Guest Digital Signature (Signature field)
- Booking Submitted On (Date-Time field, auto-captured)
- Booking ID (Auto Number field, system-generated)
- Booking Confirmation QR Code (QR/Barcode field, system-generated from Booking ID)

6. Validation & Configuration Rules

- Lookup field (Room Type Availability Lookup): connected to a master Room Inventory form; front office staff type-ahead search and select rather than typing free text.
- SubForm (Additional Guests Details): configured to allow a minimum of 0 and a maximum of 5 rows per booking.
- File Upload (ID Proof Upload): restrict accepted formats to PDF, JPG, and PNG, with a maximum size of 5 MB.
- Image (Guest Photo): restrict accepted formats to JPG and PNG only, with a maximum size of 2 MB.
- Video (Celebration Request Video): restrict maximum duration to 45 seconds and maximum size of 50 MB.
- Percent (Loyalty Discount Percent): restrict value range from 0 to 20.
- Currency (Advance Payment Paid): restrict to non-negative values only, formatted in INR (₹).
- Formula (Total Amount Payable): set as a read-only field, recalculated automatically whenever the room tariff, discount, or advance payment changes.
- Auto Number (Booking ID): configured with prefix "HTL-BKG-", the current year, and a 6-digit auto-incrementing sequence.
- QR/Barcode (Booking Confirmation QR Code): configured to auto-generate immediately after the Booking ID is assigned on submission.
- Signature (Guest Digital Signature) and Decision Box (Requires Airport Pickup) are mandatory fields; the form cannot be submitted without them.

7. Workflow Summary

- Step 1: Guest completes the existing basic-field sections (personal, contact, and room/stay preference details).
- Step 2: Front office verifies the Room Type Availability using the Lookup field, which auto-fills the current tariff and availability status.
- Step 3: For family or group bookings, additional guest details are added as rows in the Additional Guests Details SubForm.
- Step 4: Guest uploads their photo, ID proof, and optionally a celebration request video and a voice note for special arrangements.

- Step 5: Guest indicates whether airport pickup is required, and the Advance Payment Paid, Loyalty Discount, and computed Total Amount Payable are displayed.
- Step 6: Guest reviews the Hotel Policies rich text block and provides their Digital Signature to confirm acceptance of terms.
- Step 7: On Submit, Zoho Creator auto-generates the Booking ID, timestamps the Booking Submitted On field, and generates the Booking Confirmation QR Code.
- Step 8: The front office reviews the Room_Booking_Report, verifies uploaded documents, confirms the balance against Total Amount Payable, and scans the QR code for a fast, paperless check-in.

8. Expected Learning Outcomes

- Understand the purpose and correct use case for each Advanced Field in Zoho Creator.
- Learn how to configure Lookup fields for cross-form data verification.
- Practice using SubForms to capture repeating, group-based data within a single submission.
- Understand how Formula and Auto Number fields automate calculations and unique identifiers.
- Learn how to configure file-based fields (Image, File Upload, Audio, Video, Signature) with format and size restrictions.
- Understand how QR/Barcode fields can be used for real-world check-in validation and process automation.

9. Deliverable for This Lab Exercise

Students must extend their existing Room_Booking_Form in Zoho Creator to include all 17 Advanced Fields exactly as specified above, configure the listed validations and computed logic, and submit a screenshot-based lab record showing: (a) the extended form in Form Builder with all Advanced Fields, (b) a live preview of the form including the SubForm and computed Total Amount Payable, (c) at least two sample submitted records showing the auto-generated Booking ID and QR Code in the report view, and (d) a screenshot of the Room Type Availability Lookup in action.